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REVIEW

Systematic review of the effectiveness of polyurethane-coated compared with textured silicone implants in breast surgery

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Received 28 May 2015; accepted 11 January 2016

KEYWORDS

Breast implants;
Mammoplasty;
Polyurethanes;
Morbidity;
Silicone gels;
Review

Summary *Background:* Silicone gel implants are used worldwide for breast augmentation and breast reconstruction. Textured silicone implants are the most commonly placed implant, but polyurethane-coated implants are increasingly being used in an attempt to ameliorate the long-term complications associated with implant insertion.

Methods: This systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses guidelines. Electronic searches of MEDLINE, EMBASE, the Cochrane Library and www.ClinicalTrials.gov were undertaken in March 2014 using keywords.

Results: Following data extraction, 18 studies were included in the review, including four core studies of textured silicone implants and five studies reporting outcomes for polyurethane-coated silicone implants. There are no clear data reporting revision rates in patients treated with polyurethane implants. In the primary reconstructive setting, capsular contracture rates with silicone implants are 10–15% at 6 years, whilst studies of polyurethane implants report rates of 1.8–3.4%. In the primary augmentation setting, core studies show a capsular contracture rate of 2–15% at 6 years compared with 0.4–1% in polyurethane-coated implants; however, the polyurethane studies are limited by their design and poor follow-up.

Conclusions: The use of polyurethane implants should be considered a safe alternative to textured silicone implants. It is likely that an implant surface does influence short- and long-term outcomes; however, the extent of any benefit cannot be determined from the available evidence base. Future implant studies should target the short- and long-term benefits of implant surfacing by

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procedure with defined outcome measures; a head-to-head comparison would help clarify outcomes.

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Introduction

Worldwide, silicone gel- and saline-filled implants are offered for breast augmentation and post-mastectomy breast reconstruction.^{1,2} Silicone implants have been used in this setting since 1962; however, their use declined worldwide due to safety concerns in the 1990s.³ As a direct result of safety concerns, in 1988, the US Food and Drug Administration (FDA) set up a process by which all manufacturers were required to demonstrate safety and effectiveness in a premarket approval application. Silicone implants were therefore designed to meet performance and safety benchmarks before being assessed and approved by the FDA. This process has led to good-quality, long-term data being available on large cohorts of patients with silicone implants. All breast implants are associated with complications; the most frequently reported are capsular contracture, reoperation and removal. Additional complications include implant rupture, infection, asymmetry, wrinkling, scarring and pain.⁴

Polyurethane foam coatings were developed to cover silicone implants and were first used in the 1970s. Studies have indicated that polyurethane-coated silicone implants may reduce capsular contracture.^{5,6} After formation of the capsule around the implant, the polyurethane coating breaks down to become part of the capsule, and it is hypothesised that the tightening observed with traditional

silicone implants does not occur.⁷ This may be due to the polyurethane coating preventing the organised alignment of myofibroblasts, thereby interrupting the strength required for capsular contracture to occur.⁸ Following their introduction, the use of polyurethane implants spread worldwide (to >35 countries, including Europe, Australia and Asia, as well as the USA and UK).⁹

However, in 1991, a specific association was reported between polyurethane and the carcinogen 2,4-toluenediamine (TDA).¹⁰ This led to a complete prohibition of polyurethane-coated silicone implants in the USA by the FDA, which has never been lifted. In the UK, in 1992, polyurethane-coated implants were likewise prohibited although surgical use of silicone gel implants was sustained. Measures were taken in the UK to monitor the safety and effectiveness of breast implant procedures, and investigation into the carcinogenic risk of polyurethane found it to be 'small, unquantifiable'.¹¹ The CE Mark was introduced in Europe in 1996 for approval of all medical devices. For certification of breast implants, manufacturers were required to provide evidence of standards of materials, manufacturing and quality assurance. By default, this conferred approval of polyurethane-coated implants for clinical use in the UK was upheld by the Medicines and Healthcare products Regulatory Agency (MHRA) in 2005.¹²

Despite this controversy, surgical use of polyurethane-coated implants continued in other countries. As the key

concern regarding carcinogenic properties has been discharged, it would be timely to appraise research on the effectiveness of polyurethane-coated silicone implants. This systematic review aims to critically compare the reported complication rates between polyurethane-coated silicone implants and textured silicone implants to determine if an implant surface is more effective in reducing long-term morbidity.

Methods

Search strategy

This systematic review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines.¹³ Electronic searches of MEDLINE, EMBASE, the Cochrane Library and www.ClinicalTrials.gov were undertaken in March 2014 using keywords (Appendix 1). Relevant articles were independently selected from titles and abstracts and then from the full text of the manuscript. Further articles that were missed by the search strategy were identified by a manual search of the references of key articles.

Selection criteria

The search was limited to studies of adult human participants, published in English, between January 1993 and March 2014; this was to ensure that only modern designs of textured silicone and polyurethane implants were compared. Reviews, editorials, case reports, letters and comments were excluded, and only studies or case series involving at least 100 implants overall were included (Appendix 2). The quality of relevant studies was appraised using the Newcastle–Ottawa Quality Assessment Scale (NOS) for cohort or case–control studies (Table 1).¹⁴ Any disagreements were resolved by study review and discussion.

Data extraction

Data reporting the incidence of complications was extracted from all papers. The primary outcomes examined were surgical revision rate and capsular contracture rate. Secondary outcomes relating to implant safety and morbidity were also examined: implant extrusion, pain, cancer, size change, rash/irritation/inflammation, seroma, infection, asymmetry, malposition, haematoma, wrinkling/rippling and rupture.

Studies included in this review arose from cosmetic and reconstructive surgical fields. To facilitate direct comparisons of incidence following primary augmentation or primary reconstruction only, and polyurethane or silicone implants only, complication data were separated according to surgical procedure and type of implant.

Many of the studies were case series and as such did not provide a direct comparison of implant type for each type of surgical procedure. Authors of studies with mixed groups of patients were contacted directly for clarification of data sets, and these data were included when validated.

It was not possible to apply any meta-analytical or data pooling techniques to the data collected due to the heterogeneity of patient samples and follow-up, implant details and complications reported.

Results

Description of studies

A total of 40 studies fulfilled inclusion criteria, comprising seven FDA core study reports on silicone implants, 29 retrospective cohort studies and four prospective cohort studies (Figure 1).

Following data extraction, 12 retrospective cohort studies, one prospective cohort study and seven FDA core studies contributed data that could be extracted by implant type and were procedure specific (Table 1).^{7,15–33} The outcome data reported in the other 20 studies were not sufficiently specified in terms of surgical procedure and implant type to facilitate comparison.^{34–53} Eight of the studies reported data from polyurethane implants.^{7,23–26,31–33} For textured silicone implants, data from the core studies were generally used as a comparison group as these data were collected prospectively with good methodology in comparison with other included studies (see NOS scores; Table 1).

Surgical revision

Thirteen studies^{15–21,24,26–30} reported data on surgical revision, only one reporting data on polyurethane implants (Table 1).²⁴

There are no comparative data for breast reconstruction patients.

The core studies of textured silicone implants reported Kaplan–Meier risk rates for reoperation at 6 years in primary augmentation patients ranging from 18 to 28% at 6 years.^{16,17,19,20} One study reported on polyurethane implants,²⁴ this being a mixed series of mainly breast augmentations, redoaugmentations for capsular contracture⁴⁰ and a small number of breast reconstructions.²⁰ The overall rate of revision across all cohorts was just 1.2% at a mean follow-up of 7 years.²⁴ Rates quoted in this paper were per implant rather than per patient as in the core studies; this equates to a rate of approximately 2.4% per patient in this study.²⁴

Capsular contracture

The incidence of Baker III/IV capsular contracture^{7,15–21,24,27,28,30–33} was reported in 15 studies, with five studies reporting on polyurethane implants.^{7,24,30,31,33}

The Kaplan–Meier risk rate for capsular contracture in breast reconstruction patients with silicone implants ranged from 10.1% to 15.9% at 6 years,^{16–18} and was 24.6% at 10 years.²⁰ In patients receiving polyurethane implants, capsular contracture occurred in 1.8–3.4%^{30,31} of patients at 8 and 4 years, respectively, although only Grade IV capsular contracture was observed in one study.³² In both of these studies, the appearance of capsular contracture

Table 1 Characteristics of studies included.

Study	NOS ^a	Key study details ^b	Patient cohorts	Complication incidence	
FDA core studies:				Kaplan–Meier rates (% patients)	
Textured silicone implants				Surgical revision	Capsular contracture
Cunningham, 2007	9	Silicone in 1007 patients: Mentor MemoryGel implants at 3 years	Primary augmentation <i>n</i> = 551 Primary reconstruction <i>n</i> = 251	15.4% 27.0%	8.1% 8.3%
Cunningham & McCue, 2009	8	1898 silicone in 1008 patients: Mentor MemoryGel implants at 6 years	Primary augmentation <i>n</i> = 552 Primary reconstruction <i>n</i> = 251	19.4% 33.9%	9.8% 13.7%
Hammond et al., 2012	8	1831 silicone Mentor MemoryShape implants at 6-year follow-up	Primary augmentation <i>n</i> = 572 Primary reconstruction <i>n</i> = 191	18.1% 44.5%	2.4% 10.1%
Maxwell et al., 2012	8	Silicone in 941 patients Natrelle Style 410 implants at 6-year follow-up	Primary augmentation <i>n</i> = 492 Primary reconstruction <i>n</i> = 225	19.4% 43.1%	4.6% 10.7%
Spear et al., 2007	9	1348 silicone Allergan implants at 6-year follow-up	Primary augmentation <i>n</i> = 455 Primary reconstruction <i>n</i> = 98	28.0% 51.9%	14.8% 15.9%
Spear et al., 2014	9	Silicone in 715 patients Allergan implants at 10-year follow-up	Primary augmentation <i>n</i> = 455 Primary reconstruction <i>n</i> = 98	36.1% 71.5%	18.9% 24.6%
Stevens et al., 2012	9	3506 silicone in 1788 patients: Sientra implants at 5-year follow-up	Primary augmentation <i>n</i> = 1116 Primary reconstruction <i>n</i> = 225	16.6% 42.7%	8.8% 10.6%
Studies with textured silicone implants				Actual complication rates (% patients)^d	
Hvilsom et al., 2009	8	10,640 silicone in 5373 patients, at 3 and 5 years	Primary augmentation <i>n</i> = 3226 at 3 years, <i>n</i> = 1870 at 5 years	4.0%	1.2% 1.7%
Kulmala et al., 2004	7	1517 silicone; mean follow-up period of 10.9 years	Primary augmentation <i>n</i> = 685	26.0%	23.0% ^d
Medard de Chardon et al., 2010	5	Silicone in 200 patients; mean follow-up of 36 months	Primary augmentation <i>n</i> = 200	7.0%	
Stark, 1993	5	Silicone in 338 patients; mean follow-up of 5 years	Primary reconstruction <i>n</i> = 338		11.0%
Agarwal et al., 2001	3	424 silicone in 212 patients; follow-up of 3 months–7 years		Seroma 1.4%	
Studies with polyurethane-coated implants				Actual complication rates (% patients)^d	
Vazquez & Pellon, 2007	8	PU in 300 patients; 250 followed up for 10 years, 180 followed up for 15 years, remainder followed up for 5 years	Primary augmentation <i>n</i> = 300	Surgical revision	Capsular contracture 0.7%
Abel de la	5	996 PU;	Primary augmentation	<i>n/a</i> ^c	<i>n/a</i> ^c

Table 1 (continued)

Study	NOS ^a	Key study details ^b	Patient cohorts	Complication incidence
Pena-Salcedo et al., 2012		mean follow-up of 6.8 years <i>Includes mastopexy and revision patients</i>	<i>n</i> = 263 Primary reconstruction <i>n</i> = 20	
Handel et al., 2006	6	568 PU; 1537 silicone; + others; mean follow-up of 37.4 months	Primary augmentation <i>n</i> = 825 Primary reconstruction <i>n</i> = 264	19.1% ^d all; 30.0% ^d PU 15.5% 35.5%
Pompei et al., 2012	7	47 silicone; 119 PU implants	Primary reconstruction <i>n</i> = 119	0.8% PU 3.4% PU 6.3% ^d PU post-RT, 0 ^d non-RT 1.8%
Rancati et al., 2013	8	PU in 221 patients; median follow-up of 98 months	Primary reconstruction <i>n</i> = 221	
Vazquez., 1999	5	811 PU implants in 407 patients; follow-up data not available	Primary augmentation <i>n</i> = 404 Primary reconstruction <i>n</i> = 3	<i>n/a</i> ^c
Cohen et al., 1997	5	68 PU; 282 silicone in 159 patients; follow-up data not available	Primary augmentation <i>n</i> = 105 Primary reconstruction <i>n</i> = 54	Rupture 23.0% ^d all; 29.0% silicone, 0 PU
Duffy & Woods, 1994	6	681 total, inc PU and silicone; median follow-up of 49 months	Total group <i>N</i> = 200	15.0% ^d all

^a Newcastle–Ottawa Quality Assessment Scale score.^b Implant type(s) and number(s); follow-up (where provided).^c *n/a*: data not interpretable by procedure type.^d % implants, not % patients.

was restricted to those who received post-mastectomy radiotherapy.^{31,32}

In primary augmentation patients, the core studies of silicone implants reported capsular contracture rates at 6 years of 2.4–14.8%,^{16,17,19} and 18.9% at 10 years.²⁰ One series offers a single direct comparison with polyurethane implants in primary augmentation patients, reporting capsular contracture at 1%.³³ This series, however, had a limited 5-year follow-up with only 300 of 1257 patients having long-term outcomes. One further study reports on a mixed cohort of mainly breast augmentations with polyurethane implants giving a capsular contracture rate of 0.4% at 7 years.^{24,33}

Other complications

It was not possible to compare incidence of rupture, wrinkling/rippling, asymmetry, implant extrusion, cancer, breast pain and size change due to the lack of comparable data.

Haematoma

Fourteen studies provided data on the occurrence of haematoma.^{7,15,17–21,24,26,27,29,30,32,33} Four studies included data on polyurethane implants.^{24,26,30,33}

The 6-year rates for haematoma reported for primary reconstruction patients with textured silicone implants were 1.0–7.1%.^{18,19} In primary two-stage reconstructions with polyurethane implants, 0.6% of patients developed a haematoma.³⁰ In primary augmentation surgery, the haematoma rate was comparable with a rate of 1.1–2.7%^{17,18} in textured silicone implants compared with 1.2% in the polyurethane implant series.⁷

Seroma

The incidence of seroma^{7,15,17,18,20–22,24,27,30,32,33} was reported in 12 studies, including four using polyurethane implants.^{7,24,30,33}

There are no comparable data available for breast reconstruction patients.

In primary augmentation patients, the core studies reported a 6-year risk of 1.4%¹⁸ and a 10-year rate of 1.8%.²⁰ Another study²² reported a similar rate of 1.4% in primary augmentation patients with silicone implants. A single series reporting on polyurethane implants for augmentation demonstrated a similar rate of seromas occurring in 2% of primary augmentation patients.⁷

Malposition

The occurrence of malposition was reported in 10 studies,^{15,17–21,24,28,29,31} with two reporting on polyurethane implants.^{24,31}

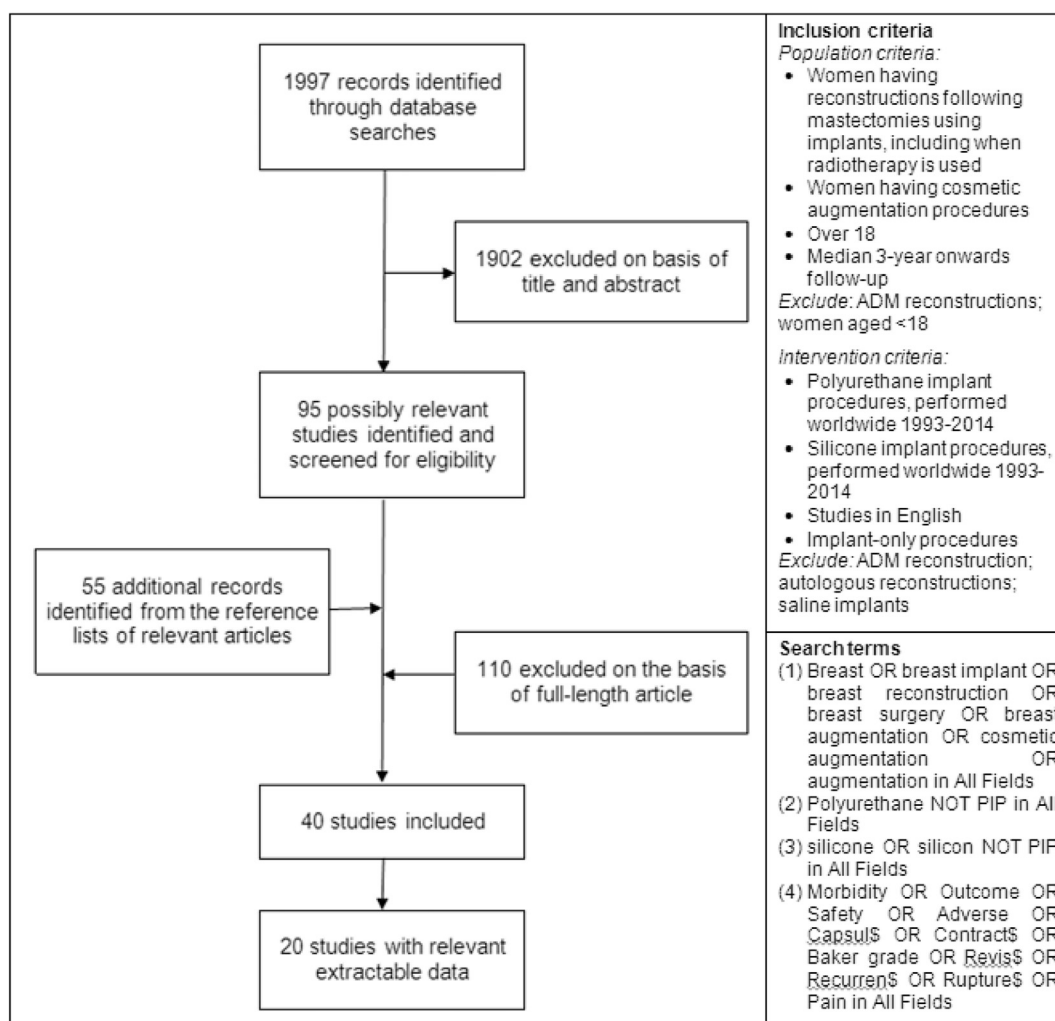


Figure 1 Study selection and search criteria.

When textured silicone implants were used for breast reconstruction, the risk of malposition at 6 years was in the range 2.1–4%^{18,19} compared to a rate of 1.8% when polyurethane was used.³¹ There are no comparable data available for breast augmentation patients.

Rash/inflammation/irritation

The incidence of rash/inflammation/irritation^{15,17,18,21,24,30,31,33} was reported in eight studies, with four being on polyurethane implants.^{24,30,31,33}

Only two of the core studies provided comparable Kaplan–Meier risk data at 6 years on the incidence of rash/inflammation/irritation in reconstruction patients, being 0.9% and 2.1%.^{17,18} By contrast, rash was reported in 4.2–5.4% of patients receiving polyurethane implants.^{30,31,33}

Only one study reported on rash in primary augmentation patients with silicone implants at 0.7–2.8%.^{18,28} Two studies including mixed cohorts (the vast majority being primary augmentations) also reported a higher incidence of rash with polyurethane implants with a rate of 4.3–4.7%.^{24,33}

Infection

Eleven studies provided data on incidence of infection^{7,15–18,21,24,26,27,30,33} including five comprising polyurethane implants.^{7,24,26,30,33}

The risk of infection with textured silicone implants at 6 years was 1.6–6.1% in primary reconstruction patients.^{16–18} By comparison, polyurethane implants are reported to have an infection rate of 0–0.4%.³⁰

In primary augmentation, rates of infection with silicone implants were 0.9% and 1.7% at 6 years.^{16–18} By comparison, a single series with a limited patient follow-up of its polyurethane implant population demonstrated a rate of 0–1%.³³

Discussion

This systematic review highlights the dichotomy in the evidence base available for textured silicone implants compared with polyurethane-coated implants. Due to the US FDA requirement for safety data on silicone implants, multiple core studies demonstrate prospectively collected

cohort data on a large number of textured silicone implants. The data from these studies allow us to study the short- and long-term outcomes of textured silicone implants as used in multiple centres, and it is therefore likely that these outcomes are realistic and achievable in other centres. By contrast, there is no similar prospective cohort study of polyurethane-coated implants. The systematic review identified eight studies reporting outcomes on polyurethane-coated implants, but only five studies presented the outcomes for a purely polyurethane implant cohort.^{7,24,30,31,33} All five studies included are single centred and retrospective and, notwithstanding their valuable findings, a limited number of small studies are available for interpretation of surgical outcomes.

The major perceived advantage of polyurethane implants is the reduced risk of capsular contracture and need for revisional surgery. The core studies of textured silicone implants demonstrate similar capsular contracture rates at 6 and 10 years and similar revision rates despite differences in the implant coating. The lack of variability in these outcomes across implant manufacturers can ensure that the true capsular contracture and revision rates lie within those reported. Capsular contracture rates reported for polyurethane implants are a factor of 3–30 times lower than textured silicone. The true rate of capsular contracture for polyurethane implants remains unclear; the data available were from single-centre and retrospective studies. Capsular contracture can only be assessed by prospective recording of capsular contracture grade in a study with defined outcome measures over a long follow-up period. The methodology from the polyurethane series does not clearly state how often these patients were assessed for this complication, and the majority of capsular contractures reported are those that required reoperation. The true rate of capsular contracture is likely to be higher than that reported.

The revision rates of surgery for patients with polyurethane implants are 15–30 times lower than the core studies. Revisional surgery occurs for many reasons, not always related to implant surface; hence, it is surprising that revision rates are very low in the polyurethane cohort, although this is reported only in two studies.^{24,30}

Many of the complications reported in the core studies may not be dependent on implant surfacing, for example, haematoma rate and infection rate. The rates of these complications would be expected to be relatively similar between the textured silicone and the polyurethane coating. Infection rates varied between 1.6% and 4.8% in breast reconstruction with textured silicone and between 0% and 0.4% with polyurethane. Similarly, haematoma rates in textured silicone implants were 1–7.1% and 0.6% in the polyurethane group. This is a surprisingly low haematoma rate compared to an accepted rate of 8% following mastectomy in a large series.⁵⁴ This further suggests that the data from these studies may not be a reflection of the true outcomes for polyurethane implants and that complications in these retrospective series are under-reported.

The systematic review included only eight polyurethane studies, thus indicating the lack of published data on the implants. It is also a reflection of the search criteria that limited the use of studies to those published from 1993 and

to studies reporting on at least 100 implants. Excluding studies and implants prior to 1993 (e.g., Castel et al., 2015) also excluded historic designs of silicone and polyurethane implants and thus attempted to make the literature relevant to modern practice.⁵⁵ Excluding smaller studies limits the risk of bias associated with these manuscripts; it would also be difficult to inform a change of practice based on a small series.

Aside from the core studies, many of the studies included have similar limitations on poor definition of outcomes, heterogeneity of groups and retrospective design. In addition, some studies reported complications on a per-implant basis and others on a per-patient basis, making comparison more difficult.

Despite the increasing use of polyurethane implants,⁵⁶ the lack of comparable reporting of complications outcomes is the most prominent inference of this review. Outcomes for textured silicone implants are well documented, but there is a lack of robust data to confirm the true nature of the complications associated with polyurethane implants. Due to the lack of prospective data on polyurethane implants, it is difficult to compare outcomes between the two implant surfaces. Polyurethane implants may be effective in reducing capsular contracture rates or revision rates; however, it is unclear from the available data whether there is a true difference between the implant types. In the absence of a multicentre, prospective study of polyurethane-coated implants, the true effect of different implant surfacing can only be studied in the long term. Certainly, further studies are required to formally assess the outcomes of polyurethane implants in both augmentation and reconstructive settings.

Conclusions

The core US FDA studies of textured silicone implants provide reliable outcome data for these implants, but there is no comparable quality evidence for polyurethane implants. This review is based on a small number of studies of polyurethane-coated devices with a limited number of participants and a considerable degree of clinical heterogeneity regarding the type of surgery and reporting of outcomes.

The use of polyurethane implants should be considered a safe alternative to textured silicone implants. Based on current evidence, any benefits of placing polyurethane implants to reduce the risk of long-term complications cannot be reliably confirmed due to a lack of evidence. It is likely that the implant surface influences short- and long-term outcomes; however, the extent of any benefit cannot be determined from the available evidence base. Future implant studies should target the short- and long-term benefits of implant surfacing by procedure with defined outcome measures; a head-to-head comparison would help clarify outcomes.

Conflicts of interest

The authors have no financial or other conflicts of interest that could influence this publication. There is no funding source. This study has not been presented or published elsewhere.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.bjps.2016.01.013>.

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